SC Honors Geometry - Nappier
Homework 10.4 – Inscribed Angles and Polygons

Name: ______________________________

Inscribed angles and arcs.

1. An inscribed angle intercepts a 120° arc. Find the angle.

2. An inscribed angle measures 36°. Find its intercepted arc.

3. In the TOP figure, $m\widehat{AB} = 96°$. Find m∠ACB.

4. Two inscribed angles intercept the same arc. One measures 39°. Find the other.

Solve for the variable.

5. A triangle is inscribed in a semicircle; its acute angles are $(4x + 6)°$ and $(5x + 3)°$. Find x and both angles.

6. An inscribed quadrilateral has opposite angles $(3x + 4)°$ and $(2x + 6)°$, and the other pair $(4y - 15)°$ and $(y + 30)°$. Find x and y.

7. In the BOTTOM figure, DF is a diameter and the two arcs above it measure $(4x + 2)°$ and $(6x - 2)°$. Find x and each arc.

Think it through.

8. An inscribed angle intercepts a semicircle. What is its measure and why?

9. A regular pentagon is inscribed in a circle. Find the arc between neighboring vertices.

10. An inscribed angle measures 54°. Find the central angle that opens onto the same arc.